EXCEL PRACTICAL ASSIGNMENT

E-Commerce Sales Analysis

1. Business Scenario

You are working as a Junior Data Analyst for an e-commerce company. The company has provided order-level sales data and a product master file. Your task is to analyze the data, perform required calculations, retrieve product information, identify important patterns, and present useful business findings using Microsoft Excel.

2. Files Provided

File	Purpose	Records
Ecommerce_Orders.xlsx	Order-level sales data	150 orders
Product_Master.xlsx	Product reference/master data	25 products

3. Instructions

- Complete the assignment using Excel only.
- Do not manually enter results where an Excel function or feature can be used.
- Use the functions and features covered in class.
- The data contains some intentional blanks, extra spaces, and inconsistent text. Handle these appropriately where required.
- For VLOOKUP tasks, use the separate Product_Master.xlsx file.
- Create additional columns in the Orders worksheet wherever a calculation or classification is required.
- Save your completed workbook with your name before submission.

Data Analysis

1. Calculate the total number of orders in the dataset.

2. Calculate the total quantity of products sold.

3. Calculate the average quantity ordered per order.

4. Find the number of orders made using each payment mode: UPI, Card, Cash, and Net Banking.

5. Find the number of orders received from each region: East, West, North, and South.

Conditional Analysis

6. Find the number of orders where Quantity is greater than or equal to 3.

7. Find the number of orders where Discount is greater than 10%.

8. Find the number of orders where Payment Mode is UPI and Region is East.

9. Find the number of orders where Quantity is greater than or equal to 3 and Discount is greater than or equal to 10%.

10. Find the number of orders handled by each salesperson.

Sales Calculations and Logical Functions

Create the required new columns in the Orders worksheet before completing these questions.

11. Create a Gross Amount column. Calculate the gross amount for every order using Quantity and Unit Price.

12. Create a Discount Amount column. Calculate the discount amount for every order.

13. Create a Net Sales column. Calculate the final sales value after deducting the discount.

14. Create a Sales Category column. Classify each order as High Value, Medium Value, or Low Value based on Net Sales: High Value = ₹50,000 or more; Medium Value = ₹20,000 or more but below ₹50,000; Low Value = below ₹20,000.

15. Create an Order Priority column. Classify an order as High when Quantity is at least 5 AND Discount is at least 15%; Medium when Quantity is at least 3 OR Discount is at least 10%; otherwise classify it as Low.

Text Functions and Data Cleaning

16. Create a new column containing only the numeric portion of Customer_ID. For example, C001 should return 001.

17. Create a new column containing only the numeric portion of Product_ID. For example, P101 should return 101.

18. Create a cleaned Salesperson column in Proper Case.

19. Create a cleaned Region column by removing unnecessary leading and trailing spaces.

VLOOKUP and Cross-File Data Retrieval

Use Product_ID in Ecommerce_Orders.xlsx to retrieve information from Product_Master.xlsx. Do not manually type the product information.

20. Using VLOOKUP, retrieve the Product Name for every order.

21. Using VLOOKUP, retrieve the Category for every order.

22. Using VLOOKUP, retrieve the Brand for every order.

23. Using VLOOKUP, retrieve the Supplier for every order.

24. Using VLOOKUP, retrieve the Product Rating for every order.

25. Retrieve Product Name, Category, Brand, Supplier, and Product Rating using the multi-value VLOOKUP technique covered in class.

Sort and Filter

26. Sort the dataset to identify the 10 orders with the highest Net Sales. Record Order ID, Customer ID, Product Name, and Net Sales.

27. Filter the dataset to display only orders where Discount is greater than or equal to 15%. Count the number of such orders.

28. Filter the dataset to display only Electronics products. Calculate the total quantity sold for these orders.

29. Filter the dataset for Region = East. Determine the number of orders, total quantity, average quantity, and total Net Sales.

Conditional Formatting

30. Apply conditional formatting to Net Sales so that high-value and low-value orders can be identified quickly.

31. Apply conditional formatting to Discount % to highlight orders with a discount of 15% or more.

32. Apply conditional formatting to Product Rating to highlight products with a rating below 4.0.

Business Questions

Use the analysis you performed above to answer the following business questions. Show the supporting calculation in Excel.

33. Which region generated the highest number of orders?

34. Which payment mode was used most frequently?

35. Which salesperson handled the highest number of orders?

36. How many orders were classified as High Value?

37. Which product category generated the highest total Net Sales?

Submission Checklist

- All required calculations are completed.
- New calculation columns are clearly named.
- VLOOKUP results have been retrieved from Product_Master.xlsx.
- Sorting and filtering tasks are completed.
- Conditional formatting has been applied.
- Business questions have been answered.
- The workbook has been saved with your name.